Aim: Write a python program to show an array operations using match-case.

Algorithm

1. Start.
2. Create an empty array.
3. Input 5 elements into the array.
4. Display the menu.
5. Read the user's choice.
6. Use `match-case`:
    a. **Case 1:** Display all array elements.
    b. **Case 2:** Insert an element at a specified position.
    c. **Case 3:** Delete an element from a specified position.
    d. **Case 4:** Search for an element and display its index.
    e. **Case 5:** Exit the program.
    f. **Default:** Display "Invalid Choice".
7. Repeat the menu until the user chooses Exit.
8. Stop.

```
# Array Operations
array = []

# Input 5 elements
print("Enter 5 array elements:")
for i in range(5):
    num = int(input(f"array[{i}] = "))
    array.append(num)

while True:
    print("\n===== ARRAY MENU =====")
    print("1. Display Array")
    print("2. Insert Element")
    print("3. Delete Element")
    print("4. Search Element")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    match choice:

        case 1:
            print("\nArray Elements:")
            for i in range(len(array)):
                print(f"array[{i}] = {array[i]}")

        case 2:
            pos = int(input("Enter position: "))
            value = int(input("Enter value: "))
            if pos >= 0 and pos <= len(array):
                array.insert(pos, value)
                print("Element inserted successfully.")
            else:
                print("Invalid position.")

        case 3:
```

```
            pos = int(input("Enter position to delete: "))
            if pos >= 0 and pos < len(array):
                del array[pos]
                print("Element deleted successfully.")
            else:
                print("Invalid position.")

        case 4:
            value = int(input("Enter element to search: "))
            if value in array:
                print("Element found at index", array.index(value))
            else:
                print("Element not found.")

        case 5:
            print("Program Ended.")
            break

        case _:
            print("Invalid Choice!")
```

Output:

Enter 5 array elements:
array[0] = 10
array[1] = 20
array[2] = 30
array[3] = 40
array[4] = 50

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

Enter your choice: 1

Array Elements:
array[0] = 10
array[1] = 20
array[2] = 30
array[3] = 40
array[4] = 50

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

```
Enter your choice: 2
Enter position: 2
Enter value: 25

Element inserted successfully.

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

Enter your choice: 1

Array Elements:
array[0] = 10
array[1] = 20
array[2] = 25
array[3] = 30
array[4] = 40
array[5] = 50

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

Enter your choice: 3
Enter position to delete: 4

Element deleted successfully.

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

Enter your choice: 1

Array Elements:
array[0] = 10
array[1] = 20
array[2] = 25
array[3] = 30
array[4] = 50

===== ARRAY MENU =====
1. Display Array
2. Insert Element
```

3. Delete Element
4. Search Element
5. Exit

Enter your choice: 4
Enter element to search: 25

Element found at index 2

===== ARRAY MENU =====
1. Display Array
2. Insert Element
3. Delete Element
4. Search Element
5. Exit

Enter your choice: 5

Program Ended.